

CS 101 Lab 6 Revision Thu

Terrain Smoothing Filter

Create a new file named `Lab06_RevThu.java` and copy your base code into it.

You need to smooth out the noisy terrain data. Implement a static method `int[][] smoothTerrain(int[][] grid)` that creates and returns a *new* 2D array of the same dimensions.

For every cell, its new value will be the integer average of itself and its valid immediate neighbors (up, down, left, right).

Hint: You must calculate these averages using the values from the *original* grid. Do not overwrite the original grid while calculating, or your math for subsequent cells will be wrong.

After displaying the original peaks, call your method and print the newly smoothed terrain map.

The following table:

Example 1: <pre>Enter number of rows: 3 Enter number of columns: 3 Enter elevations: 10 0 10 0 20 0 10 0 10 ... (Peak output omitted) Smoothed Terrain Map: 3 8 3 8 4 8 3 8 3</pre>	Example 2: <pre>Enter number of rows: 2 Enter number of columns: 3 Enter elevations: 5 5 5 5 10 5 ... (Peak output omitted) Smoothed Terrain Map: 5 6 5 6 6 6</pre>
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